At the physical layer, binary expressions (that is, a series of 1s and 0s) represent data. A binary expression is created using bits, where a bit is a single 1 or a single 0. At upper layers, however, bits are grouped together, into what is known as a ***protocol data unit (PDU)*** or a *data service unit*.

Engineers tend to use the term *packet* generically to refer to these PDUs. However, PDUs might have an added name, depending on their OSI layer. Figure 1-3 illustrates these PDU names. A common memory aid for these PDUs is ***S**ome **P**eople **F**ear **B**irthdays*, where the *S* in *Some* reminds us of the *S* in *Segments*. The *P* in *People* reminds us of the *P* in *Packets*, and the *F* in *Fear* reflects the *F* in *Frames*. Finally, the *B* in *Birthdays* reminds us of the *B* in *Bits*.

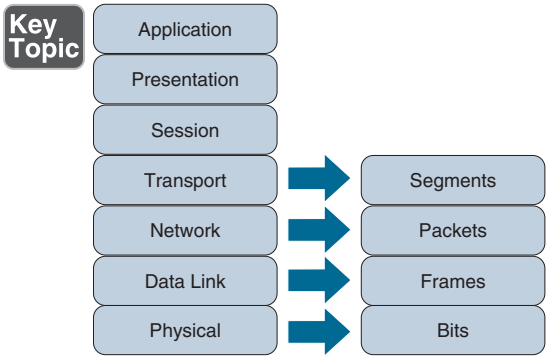


Figure 1-3 PDU Names

Layer 1: The Physical Layer

The concern of the ***physical layer***, as shown in Figure 1-4, is the transmission of bits on the network along with the physical and electrical characteristics of the network.

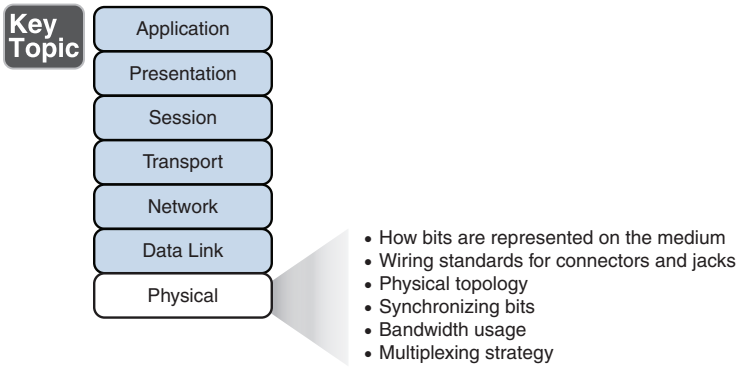


Figure 1-4 Layer 1: The Physical Layer

The physical layer defines the following:

- **How to represent bits on the medium:** Data on a computer network is represented as a binary expression. Chapter 7, "IPv4 Addressing," discusses binary in much more detail. Electrical voltage (on copper wiring) or light (carried via fiber-optic cabling) can represent these 1s and 0s.

  For example, the presence or absence of voltage on a wire portrays a binary 1 or a binary 0, respectively, as illustrated in Figure 1-5. Similarly, the presence or absence of light on a fiber-optic cable renders a 1 or 0 in binary. This type of approach is called ***current state modulation***.

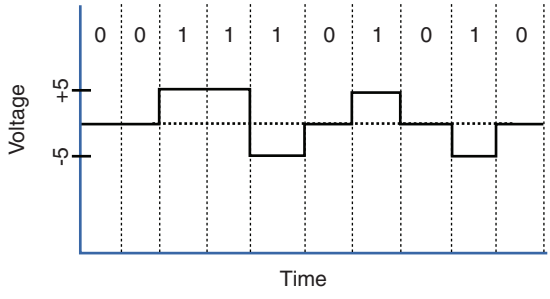


Figure 1-5 Current State Modulation

An alternative approach to portraying binary data is ***state transition modulation***, as shown in Figure 1-6, where the transition between voltages or the presence of light shows a binary value.

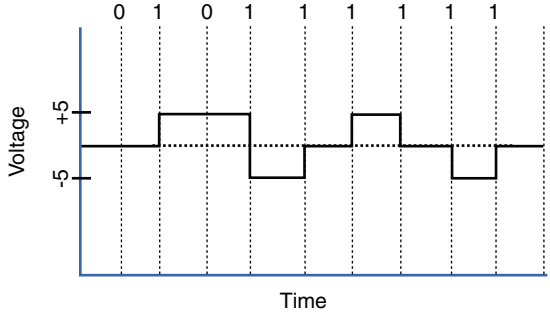


Figure 1-6 Transition Modulation

NOTE Other modulation types you might be familiar with from radio include amplitude modulation (AM) and frequency modulation (FM). AM uses a variation in a waveform's amplitude (that is, signal strength) to portray the original signal. FM uses a variation in frequency to stand for the original signal.

- **Wiring standards for connectors and jacks:** Chapter 5, "Transmission Media and Transceivers," describes several standards for network connectors. For example, the TIA/EIA-568-B standard describes how to wire an RJ-45 connector for use on a 100BASE-TX Ethernet network, as shown in Figure 1-7.

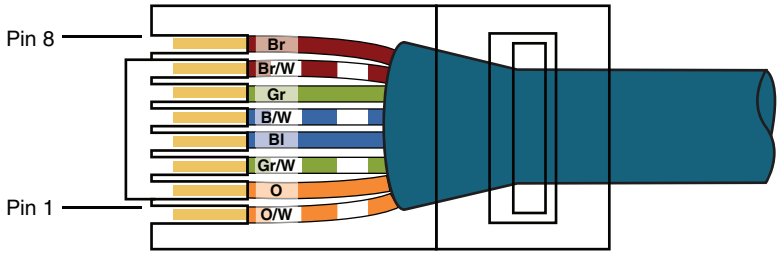


Figure 1-7 TIA/EIA-568-B Wiring Standard for an RJ-45 Connector

- **Physical topology:** Layer 1 devices view a network as a physical topology (as opposed to a logical topology). Examples of a physical topology include star and hub-and-spoke topologies, as described in Chapter 6, "Network Topologies, Architectures, and Types."
- **Synchronizing bits:** For two networked devices to successfully communicate at the physical layer, they must agree on when one bit stops and another bit starts. Specifically, the devices need a method to synchronize the bits. Two basic approaches to bit synchronization are *asynchronous* and *synchronous* synchronization:
  - **Asynchronous:** With this approach, a sender states that it is about to start transmitting by sending a start bit to the receiver. When the receiver sees this, it starts its own internal clock to measure the next bits. After the sender transmits its data, it sends a stop bit to say that it has finished its transmission.
  - **Synchronous:** This approach synchronizes the internal clocks of the sender and the receiver to ensure that they agree on when bits begin and end. A common approach to make this synchronization happen is to use an external clock (for example, a clock provided by a service provider). The sender and receiver then reference this external clock.
- **Bandwidth usage:** The two fundamental approaches to bandwidth usage on a network are *broadband* and *baseband*:
  - **Broadband:** Broadband technologies divide the bandwidth available on a medium (for example, copper or fiber-optic cabling) into different

channels. A sender can then transmit different communication streams over the various channels. For example, consider frequency-division multiplexing (FDM) used by a cable modem. Specifically, a cable modem uses certain ranges of frequencies on the cable coming into your home from the local cable company to carry incoming data, another range of frequencies for outgoing data, and several other frequency ranges for various TV stations.

- **Baseband:** Baseband technologies use all the available frequencies on a medium to send data. Ethernet is an example of a networking technology that uses baseband.

- **Multiplexing strategy:** Multiplexing allows multiple communications sessions to share the same physical medium. Cable TV, as previously mentioned, allows you to receive multiple channels over a single physical medium (for example, a coaxial cable plugged into the back of your television). Here are some of the most common approaches to multiplexing:

  - ***Time-division multiplexing (TDM)*:** TDM supports different communication sessions (for example, different telephone conversations in a telephony network) on the same physical medium by causing the sessions to take turns. For a brief period, defined as a *time slot*, data from the first session is sent, followed by data from the second session. This continues until all sessions have had a turn, and the process repeats.

  - **Statistical time-division multiplexing (StatTDM):** A downside to TDM is that each communication session receives its own time slot, even if one of the sessions does not have any data to send at the moment. To make more efficient use of available bandwidth, StatTDM dynamically assigns time slots to communications sessions on an as-needed basis.

  - **Frequency-division multiplexing (FDM):** FDM divides a medium's frequency range into channels, and different communication sessions send their data over different channels. As previously described, this approach to bandwidth usage is called *broadband*.

  - **Orthogonal frequency-division multiplexing (OFDM):** OFDM encodes digital data onto multiple carrier frequencies. OFDM is very popular today and is used in wideband digital communication. This makes OFDM useful in applications such as digital television and audio broadcasting, DSL Internet access, wireless networks, powerline networks, and 4G/5G mobile communications.

- **Orthogonal frequency-division multiple access (OFDMA):** OFDMA is a multiuser version of the popular OFDM digital modulation scheme. It divides a wireless channel into subchannels to allow simultaneous data transmission from multiple users, enhancing network efficiency and reducing latency.

Examples of devices defined by physical layer standards include legacy hubs, wireless access points, and network cabling.

NOTE Hubs are not used in modern computer networks. So why are we even bothering to mention them? Well, they really did help give rise to our modern switches. A hub interconnects PCs in a LAN; it is considered a physical layer device because it takes bits coming in on one port and retransmits those bits out all other ports. At no point does the hub learn any addressing information from the data as our modern switches do.

Layer 2: The Data Link Layer

The ***data link layer*** is concerned with the following:

- Packaging data into frames and transmitting those frames on the network
- Ensuring that frames do not exceed the ***maximum transmission unit (MTU)*** of the physical media
- Performing error detection/correction
- Uniquely finding network devices with addresses
- Handling flow control

NOTE Network interfaces use the MTU to define the largest packet size the interface will forward. For example, a 1500-byte packet could not be forwarded via a router interface with an MTU of 1470 bytes. In Chapter 22, "Troubleshoot Common Cabling Problems," you will learn about *giants* and *runts*—packets that are too large or too small for the network on which they reside.

Data link layer processes, collectively referred to as *data link control* (*DLC*), are illustrated in Figure 1-8.

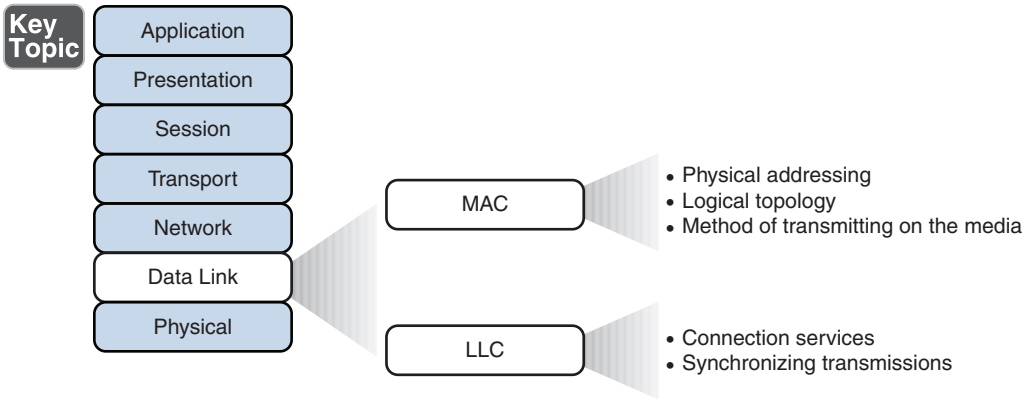


Figure 1-8 Layer 2: The Data Link Layer

In fact, the data link layer is distinct from the other layers in that it has two sublayers: MAC and LLC.

Media Access Control

Characteristics of the Media Access Control (MAC) sublayer of the data link layer include the following:

- **Physical addressing:** A common example of a Layer 2 address is a MAC address, which is a 48-bit address assigned to a device's network interface card (NIC). MAC addresses are written in hexadecimal notation (for example, 58:55:ca:eb:27:83). The first 24 bits of the 48-bit address are the *vendor code*. The IEEE Registration Authority assigns a manufacturer one or more unique vendor codes. You can use the list of vendor codes at https://standards-oui.ieee.org/oui/oui.txt to identify the manufacturer of a networking device, based on the first half of the device's MAC address. The last 24 bits of a MAC address are assigned by the manufacturer, and they act as a serial number for the device. No two MAC addresses in the world should have the same value.
- **Logical topology:** Layer 2 devices view a network as a logical (as opposed to physical) topology. Chapter 6 has lots more to say regarding physical and logical topologies.
- **Method of transmitting on the media:** With several devices connected to a network, there needs to be some strategy for deciding when a device sends on the media. Otherwise, multiple devices might send at the same time and interfere with one another's transmissions.

Logical Link Control

Characteristics of the Logical Link Control (LLC) sublayer of the data link layer include the following:

- **Connection services:** When a device on a network receives a message from another device on the network, that recipient device can give feedback to the sender in the form of an acknowledgment message. The two main functions provided by these acknowledgment messages are as follows:
  - **Flow control:** Limits the amount of data a sender can send at one time; this prevents the sender from overwhelming the receiver with too much information.
  - **Error control:** Allows the recipient of data to let the sender know whether the expected data frame was not received or whether it was received but is corrupted. The recipient figures out whether the data frame is corrupt by mathematically calculating a checksum of the data received. If the calculated checksum does not match the checksum received with the data frame, the recipient of the data draws the conclusion that the data frame is corrupted and can then notify the sender via an acknowledgment message.
- **Synchronizing transmissions:** Senders and receivers of data frames need to coordinate when a data frame is being transmitted and should be received. The three methods of performing this synchronization are detailed here:
  - **Isochronous:** With isochronous transmission, network devices look to a common device in the network as a clock source, which creates fixed-length time slots. Network devices can determine how much free space, if any, is available within a time slot and then insert data into an available time slot. A time slot can accommodate more than one data frame. Isochronous transmission does not need to provide clocking at the beginning of a data string (as does synchronous transmission) or for every data frame (as does asynchronous transmission). As a result, isochronous transmission uses little overhead compared to asynchronous or synchronous transmission methods.
  - **Asynchronous:** With asynchronous transmission, network devices reference their own internal clocks, and network devices do not need to synchronize their clocks. Instead, the sender places a start bit at the beginning of each data frame and a stop bit at the end of each data frame. These start and stop bits tell the receiver when to monitor the medium for the presence of bits.

An additional bit, called the *parity bit*, might also be added to the end of each byte in a frame to detect an error in the frame. For example, if even parity error detection (as opposed to odd parity error detection) is used, the parity bit (with a value of either 0 or 1) would be added to the end of a byte, causing the total number of 1s in the data frame to be an even number. If the receiver of a byte is configured for even parity error detection and receives a byte where the total number of bits (including the parity bit) is even, the receiver can conclude that the byte was not corrupted during transmission.

NOTE Using a parity bit to detect errors might not be effective if a byte has more than one error (that is, if more than one bit has been changed from its original value).

- **Synchronous:** With synchronous transmission, two network devices that want to communicate between themselves must agree on a clocking method to show the beginning and ending of data frames. One approach to providing this clocking is to use a separate communications channel over which a clock signal is sent. Another approach relies on specific bit combinations or control characters to indicate the beginning of a frame or a byte of data.

  Like asynchronous transmissions, synchronous transmissions can perform error detection. However, rather than using parity bits, synchronous communication runs a mathematical algorithm on the data to create a ***cyclic redundancy check (CRC)***. If the sender and the receiver calculate the same CRC value for the same chunk of data, the receiver can conclude that the data was not corrupted during transmission.

Examples of devices defined by data link layer standards include switches, bridges, and NICs.

NOTE NICs are not entirely defined at the data link layer because they are partially based on physical layer standards, such as a NIC's network connector.

Layer 3: The Network Layer

The ***network layer***, as shown in Figure 1-9, is primarily concerned with forwarding data based on logical addresses.

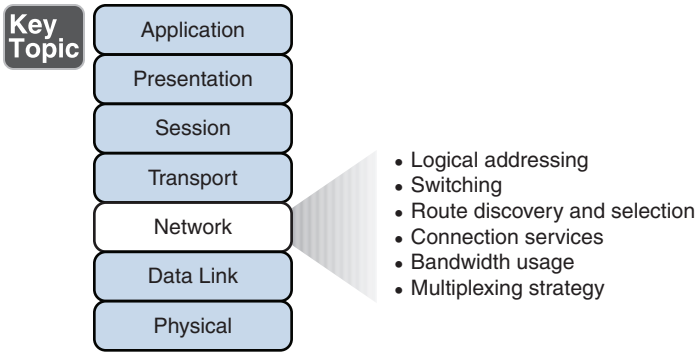


Figure 1-9 Layer 3: The Network Layer

Although many network administrators think of routing and IP addressing when they hear about the network layer, this layer is actually responsible for a variety of tasks:

- **Logical addressing:** Whereas the data link layer uses physical addresses to make forwarding decisions, the network layer uses logical addressing to make forwarding decisions. By far the most widely deployed routed protocol is Internet Protocol (IP). Chapter 7 discusses IP addressing in detail.
- **Switching:** Engineers often associate the term *switching* with Layer 2 technologies; however, the concept of switching also exists at Layer 3. Switching, at its essence, is making decisions about how data should be forwarded. At Layer 3, three common switching techniques exist:
  - **Packet switching:** With packet switching, a data stream is divided into packets. Each packet has a Layer 3 header that includes source and destination Layer 3 addresses. Another term for packet switching is *routing*, which is discussed in more detail in Chapter 9, "Routing Technologies."
  - **Circuit switching:** Circuit switching dynamically brings up a dedicated communication link between two parties for those parties to communicate.

    As a simple example of circuit switching, think of making a phone call from your home to a business. In fact, let's go "old school" and pretend you have a traditional landline servicing your phone, the telephone company's switching equipment interconnects your home phone with the phone system of the business you are calling. This interconnection (that is, *circuit*) exists only for the duration of the phone call.
  - **Message switching:** Unlike packet switching and circuit switching technologies, message switching is usually not well suited for real-time

applications because of the delay involved. Specifically, with message switching, a data stream is divided into messages. Each message is tagged with a destination address, and the messages travel from one network device to another network device on the way to their destination. Because these devices might briefly store the messages before forwarding them, a network using message switching is sometimes called a *store-and-forward* network. Metaphorically, you could visualize message switching like routing an email message, where the email message might be briefly stored on an email server before being forwarded to the recipient.

- **Route discovery and selection:** Because Layer 3 devices make forwarding decisions based on logical network addresses, a Layer 3 device might need to know how to reach various network addresses. For example, a common Layer 3 device is a router. A router can maintain a routing table indicating how to forward a packet based on the packet's destination network address.

  A router can have its routing table populated via manual configuration (that is, by entering static routes), via a dynamic routing protocol (for example, OSPF or EIGRP), or simply by being directly connected to certain networks.

NOTE Routing protocols are discussed in Chapter 9.

- **Connection services:** Just as the data link layer offers connection services for flow control and error control, connection services also exist at the network layer. Connection services at the network layer can improve the communication reliability if the data link's LLC sublayer is not performing connection services. The following functions are performed by connection services at the network layer:
  - **Flow control (also known as congestion control):** Helps prevent a sender from sending data more rapidly than the receiver is capable of receiving it.
  - **Packet reordering:** Allows packets to be placed in the proper sequence as they are sent to the receiver. This might be necessary because some networks support load balancing, where multiple links are used to send packets between two devices. Because multiple links exist, packets might arrive out of order.

Examples of devices found at the network layer include routers and multilayer switches. The most common Layer 3 protocol in use, and the protocol on which the Internet is based, is IPv4. However, IPv6 is beginning to be more common on networks today.

Layer 4: The Transport Layer

The ***transport layer***, as shown in Figure 1-10, acts as a dividing line between the upper layers and lower layers of the OSI model. Specifically, messages are taken from upper layers (Layers 5–7) and are encapsulated into segments for transmission to the lower layers (Layers 1–3). Similarly, data streams coming from lower layers are decapsulated and sent to Layer 5 (the session layer), or some other upper layer, depending on the protocol.

Two common transport layer protocols are TCP and UDP:

- ***Transmission Control Protocol (TCP)*:** TCP is a connection-oriented transport protocol. Connection-oriented transport protocols offer reliable transport, in that if a segment is dropped, the sender can detect the drop and retransmit the dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can decide which segments were successfully received and which segments need to be transmitted again.
- ***User Datagram Protocol (UDP)*:** UDP is a connectionless transport protocol. Connectionless transport protocols offer unreliable transport, in that if a segment is dropped, the sender is unaware of the drop, and no retransmission occurs.

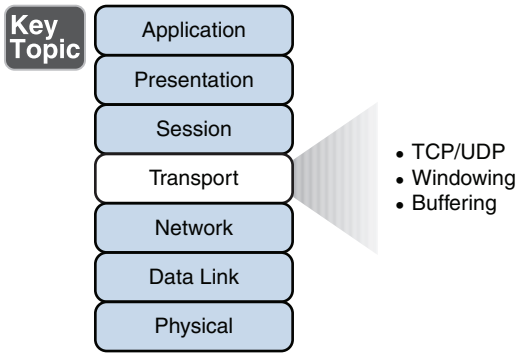


Figure 1-10 Layer 4: The Transport Layer

Just as Layer 2 and Layer 3 offer flow control services, flow control services also exist at Layer 4. Two common flow control approaches at Layer 4 are windowing and buffering:

- **Windowing:** TCP communication uses windowing, in that one or more segments are sent at one time, and a receiver can attest to the receipt of all the segments in a window with a single acknowledgment. In some cases, as

illustrated in Figure 1-11, TCP uses a sliding window, where the window size begins with one segment. If there is a successful acknowledgment of that one segment (that is, the receiver sends an acknowledgment asking for the next segment), the window size doubles to two segments. Upon successful receipt of those two segments, the next window holds four segments. This exponential increase in window size continues until the receiver does not acknowledge successful receipt of all segments within a certain amount of time—known as the *round-trip time* (RTT), which is sometimes called *real transfer time*—or until a configured maximum window size is reached.

- **Buffering:** With buffering, a device (for example, a router) uses a chunk of memory (sometimes called a *buffer* or a *queue*) to store segments if bandwidth is not available to send those segments. A queue has finite capacity, however, and can overflow (that is, drop segments) in the event of sustained network congestion.

Key Topic

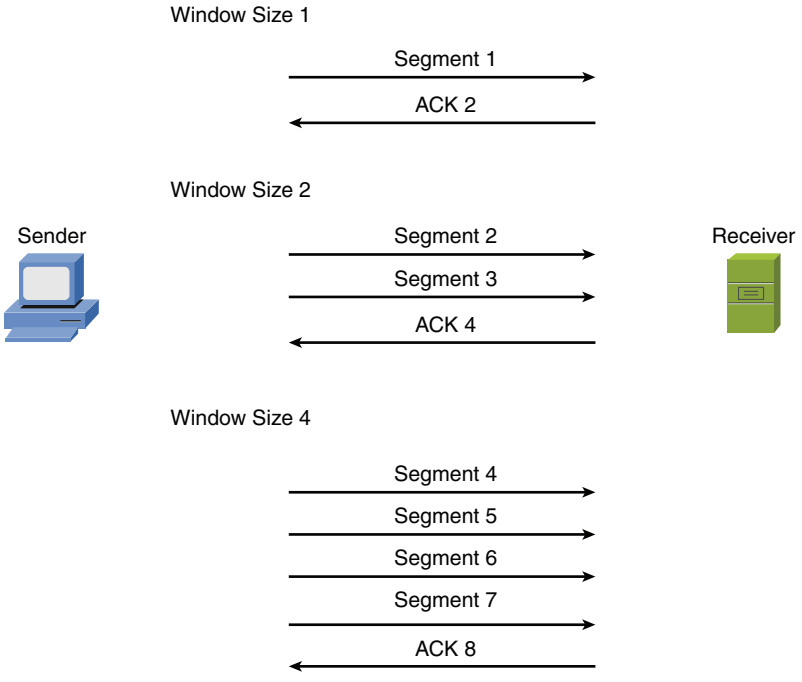


Figure 1-11 TCP Sliding Window

In addition to TCP and UDP, Internet Control Message Protocol (ICMP) is another transport layer protocol you are likely to meet. ICMP is used by utilities

such as ping and traceroute, which are discussed in Chapter 25, "Network Troubleshooting Tools."

Layer 5: The Session Layer

The ***session layer***, as shown in Figure 1-12, is responsible for setting up, maintaining, and tearing down sessions. You can think of a session as a conversation that needs to be treated separately from other sessions to avoid the intermingling of data from different conversations.

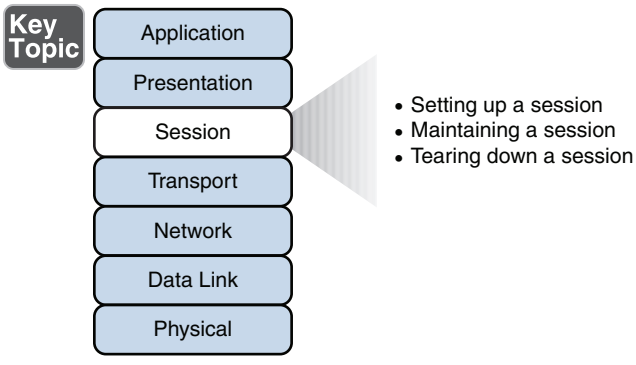


Figure 1-12 Layer 5: The Session Layer

Here is a detailed look at the functions of the session layer:

- **Setting up a session:** Examples of the procedures involved in setting up a session include the following:
  - Checking user credentials (for example, username and password)
  - Assigning numbers to a session's communication flows to uniquely find each one
  - Negotiating services needed during the session
  - Negotiating which device begins sending data
- **Maintaining a session:** Examples of the procedures involved in supporting a session include the following:
  - Transferring data
  - Reestablishing a disconnected session
  - Acknowledging receipt of data

- **Tearing down a session:** A session can be disconnected based on agreement of the devices in the session. Alternatively, a session might be torn down because one party disconnects (either intentionally or because of an error condition). If one party disconnects, the other party can detect a loss of communication with that party and tear down its side of the session.

Session Initiation Protocol (SIP) is an example of a session layer protocol, which can help set up, support, and tear down a voice or video connection. Keep in mind, however, that not every network application neatly maps directly to all seven layers of the OSI model.

Layer 6: The Presentation Layer

The ***presentation layer***, as shown in Figure 1-13, formats the data being exchanged and secures that data with encryption.

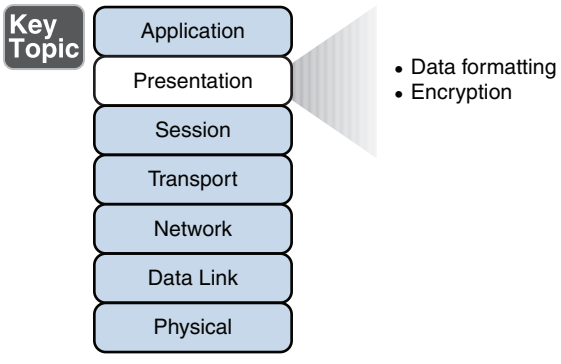


Figure 1-13 Layer 6: The Presentation Layer

The following list describes the functions involved in data formatting and encryption in more detail:

- **Data formatting:** As an example of how the presentation layer handles data formatting, consider how text is formatted. Some applications might format text using American Standard Code for Information Interchange (ASCII), while other applications might format text using Extended Binary Coded Decimal Interchange Code (EBCDIC). The presentation layer handles formatting the text (or other types of data, such as multimedia or graphics files) in a format that allows compatibility between the communicating devices.

- **Encryption:** Imagine that you are sending sensitive information over a network (for example, your credit card number or bank password). If a malicious user were to intercept your transmission, they might be able to obtain this sensitive information. To add a layer of security for such transmissions, encryption can be used to scramble (encrypt) the data in such a way that if the data were intercepted, a third party would not be able to unscramble (decrypt) it. However, the intended recipient would be able to decrypt the transmission.

Encryption is discussed in detail in Chapter 18, "Network Security Concepts."

Layer 7: The Application Layer

The ***application layer***, as shown in Figure 1-14, gives application services to a network. An important (and often-misunderstood) concept is that end-user applications (such as Microsoft Word) live at the application layer. Instead, the application layer supports services used by end-user applications. For example, email is an application layer service that does exist at the application layer, whereas Microsoft Outlook (an example of an email client) is an end-user application that does not live at the application layer. Another function of the application layer is advertising available services.

Key Topic

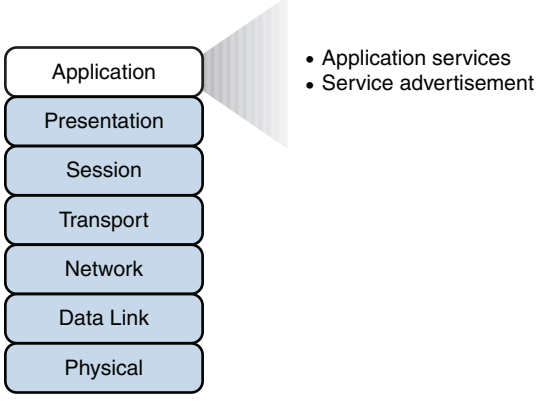


Figure 1-14 Layer 7: The Application Layer

The following list describes the functions of the application layer in more detail:

- **Application services:** Examples of the application services living at the application layer include file sharing and email.
- **Service advertisement:** Some applications' services (for example, some networked printers) periodically send out advertisements, making their availability known to other devices on the network. Other services, however,

register themselves and their services with a centralized directory (for example, Microsoft Active Directory), which can be queried by other network devices seeking such services.

Recall that even though the application layer is numbered as Layer 7, it is at the top of the OSI stack because its networking functions are closest to the end user.

The TCP/IP Stack

The ISO developed the OSI reference model to be generic, in terms of what protocols and technologies could be categorized by the model. However, most of the traffic on the Internet (and traffic on corporate networks) is based on the TCP/IP protocol suite. Therefore, a more relevant model for many network designers and administrators to reference is a model developed by the U.S. Department of Defense (DoD). This model is known as the *DoD model*, or the ***TCP/IP stack***.

NOTE An older protocol known as Network Control Protocol (NCP) was similar to TCP/IP. NCP was used on ARPANET (the predecessor to the Internet), and it provided features like those offered by the TCP/IP suite of protocols on the Internet, although they were not as robust.

Layers of the TCP/IP Stack

The TCP/IP stack has only four defined layers, as opposed to the seven layers of the OSI model. Figure 1-15 contrasts these two models.

Key Topic

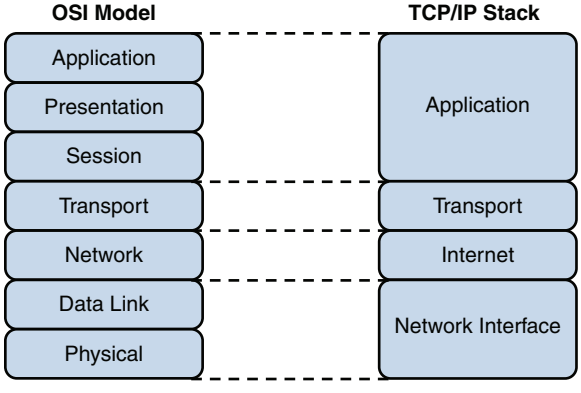


Figure 1-15 TCP/IP Stack